

Polynomial Multiplication & Division

Algebra 1 (US Common Core) | Worksheet 2 – Comprehensive Practice

1 Polynomial Multiplication

Definition

Multiplying polynomials involves applying the distributive property to multiply every term in the first polynomial by every term in the second, then combining like terms using exponent rules ($x^a \cdot x^b = x^{a+b}$).

Worked Example (Hard): Multiply and simplify $(3x^2 - 4x + 5)(2x^2 + 5x - 3)$

$$\begin{aligned} &= 3x^2(2x^2 + 5x - 3) - 4x(2x^2 + 5x - 3) + 5(2x^2 + 5x - 3) \\ &= (6x^4 + 15x^3 - 9x^2) + (-8x^3 - 20x^2 + 12x) + (10x^2 + 25x - 15) \\ &= 6x^4 + (15 - 8)x^3 + (-9 - 20 + 10)x^2 + (12 + 25)x - 15 \\ &= 6x^4 + 7x^3 - 19x^2 + 37x - 15 \end{aligned}$$

Homework Problems

1. $(4x^2 - 7)(3x^2 + 2x - 5)$
2. $(2x - 3y)^3$
3. $(5x^2 - 2x + 4)(3x^2 + 4x - 1) - (x^3 - 6x + 2)$
4. $(x + 2y - z)^2$
5. $(3x^3 - 5x^2 + 2x - 1)(2x^2 - 4)$
6. $(x^2 - 3x + 9)(x + 3)(x^2 - 9)$
7. $(2x^2 + y^2 - 3xy)(4x^2 - 2xy + y^2)$
8. $(3x - 1)^4$
9. $(a^2 - ab + b^2)(a^2 + ab + b^2)$
10. $(x^4 - 2x^2 + 3)(3x^2 - 5)$
11. $(2x^2 + 3x - 4)^2$
12. $(x + y + z)(x - y + z)$
13. $(5x^3 - 2x^2 + 4x - 7)(3x - 2)$

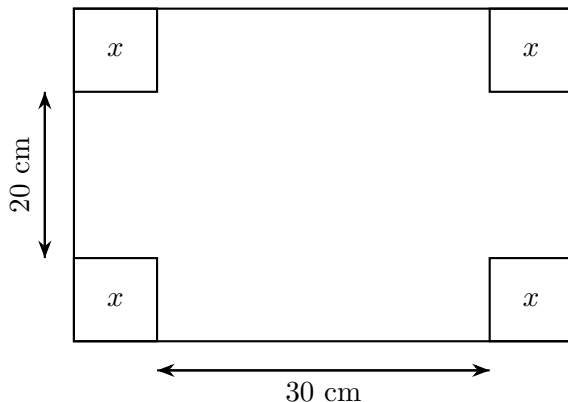
2 Word Problems

Definition

Algebraic word problems translate real-world geometric, motion, or financial scenarios into polynomial expressions and equations.

Worked Example (Hard)

A box with an open top is constructed from a $20 \text{ cm} \times 30 \text{ cm}$ rectangular piece of cardboard by cutting square corners of side length $x \text{ cm}$ and folding up the sides. Express the volume $V(x)$ and find the volume when $x = 3 \text{ cm}$.



Dimensions: Length = $30 - 2x$, Width = $20 - 2x$, Height = x .

$$V(x) = x(30 - 2x)(20 - 2x) = x(600 - 100x + 4x^2) = 4x^3 - 100x^2 + 600x.$$

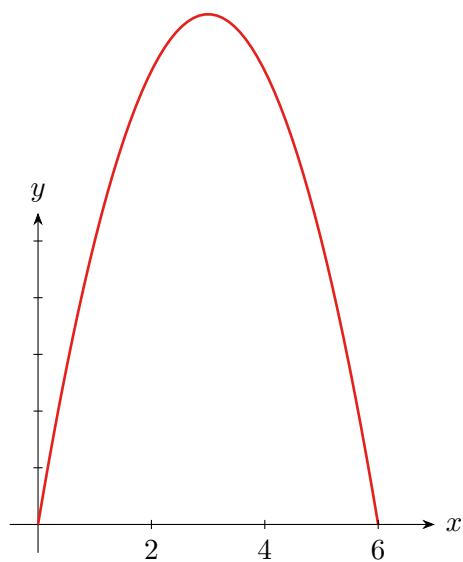
For $x = 3$: $V(3) = 4(3)^3 - 100(3)^2 + 600(3) = 108 - 900 + 1800 = 1008 \text{ cm}^3$.

Homework Problems

14. A rectangle's length is $3x - 2$ and width is $2x + 5$. If a square of side $x + 1$ is removed from its center, write a polynomial for the remaining area.
15. The height of a projectile is $h(t) = -16t^2 + 80t + 96$. Factor completely to find when it hits the ground ($h = 0$).
16. An open-top box is made from a 16×24 inch sheet by cutting squares of side x from each corner. Express the volume polynomial in standard form.
17. A garden of area $A(x) = 6x^2 + 11x - 10$ has width $2x + 5$. Find a polynomial for its length.
18. A cube's side length is increased by 2 units, increasing its volume by 98 cubic units. Find the original side length.
19. Revenue $R(x) = 150x - 2x^2$, cost $C(x) = 30x + 400$. Write and simplify the profit polynomial $P(x)$, then find the maximum profit.
20. A rectangular pool ($20 \text{ ft} \times 30 \text{ ft}$) is surrounded by a uniform walkway of width x . Write a polynomial for the total area including the walkway.
21. Three consecutive odd integers have a product that exceeds 15 times their sum by 80. Set up the polynomial equation.
22. The product of two numbers is $2x^3 + 5x^2 - 4x - 3$. If one number is $x + 3$, find the other.
23. A cylinder has radius $r = x + 2$ and height $h = 2x - 1$. Express its total surface area $A(x)$ in terms of π .

Problem 24

The path of a parabolic arch is $y = -x^2 + 6x$. Find the maximum height and the width at the base.



3 Division with Remainder Zero

Definition

Polynomial division with a remainder of zero occurs when the divisor is an exact factor of the dividend. By the Factor Theorem, if $P(c) = 0$, then $(x - c)$ divides $P(x)$ evenly.

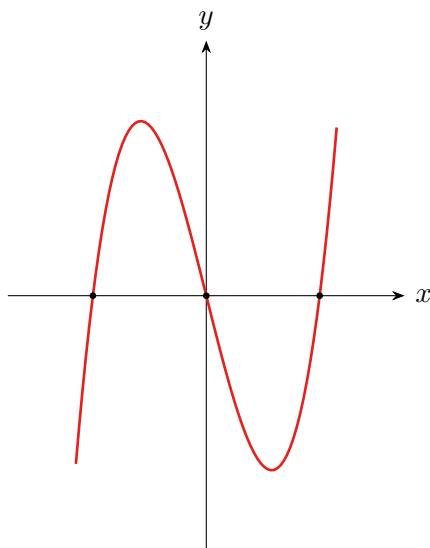
Worked Example (Hard): Divide $(2x^4 - 5x^3 - 11x^2 + 20x + 12)$ by $(2x + 1)$

$$\begin{array}{r|l}
 & x^3 - 3x^2 - 4x + 12 \\
 2x + 1 & 2x^4 - 5x^3 - 11x^2 + 20x + 12 \\
 & -(2x^4 + x^3) \\
 \hline
 & -6x^3 - 11x^2 \\
 & -(-6x^3 - 3x^2) \\
 \hline
 & -8x^2 + 20x \\
 & -(-8x^2 - 4x) \\
 \hline
 & 24x + 12 \\
 & -(24x + 12) \\
 \hline
 & 0
 \end{array}$$

Quotient: $x^3 - 3x^2 - 4x + 12$ (Remainder = 0).

Homework Problems

25. Divide $(3x^3 - 2x^2 - 7x - 2) \div (3x + 1)$
26. Divide $(x^4 - 16) \div (x - 2)$
27. Divide $(2x^4 + 3x^3 - 8x^2 - 9x + 6) \div (2x^2 - 3)$
28. Divide $(6x^3 + 17x^2 + 22x + 15) \div (2x + 3)$
29. Find the quotient $(x^5 + 32) \div (x + 2)$
30. Divide $(4x^4 - 4x^3 - 11x^2 + 6x + 6) \div (2x^2 - 3)$
31. Divide $(3x^4 - 10x^3 + 12x^2 - 11x + 6) \div (x^2 - 2x + 3)$
32. Divide $(2x^3 - 9x^2 + 13x - 6) \div (2x - 3)$
33. Divide $(x^4 + 2x^3 - 13x^2 - 14x + 24) \div (x^2 - x - 6)$
34. Find k such that $(x + 3)$ divides $(2x^3 + 5x^2 - kx + 6)$ with zero remainder.
35. Divide $(6x^4 - x^3 - 13x^2 + 4x + 4) \div (2x^2 + x - 2)$
36. Graphically verify that $f(x) = x^3 - 4x$ has roots at $x = -2, 0, 2$ by performing polynomial division for each linear factor.



More Homework Problems

37. Divide $(x^4 - 3x^3 + 6x^2 - 12x + 8) \div (x^2 + 4)$

38. Divide $(8x^3 + 27) \div (2x + 3)$

39. Divide $(3x^3 - 4x^2 + 5x - 2) \div (x - \frac{1}{3})$

40. Show that $(x^2 - 2)$ is a factor of $(x^4 - x^3 - 2x^2 + 2x)$ by completing the division.

4 Answer Key

1. $12x^4 + 8x^3 - 41x^2 - 14x + 35$
2. $8x^3 - 36x^2y + 54xy^2 - 27y^3$
3. $15x^4 + 13x^3 - 3x^2 + 24x - 6$
4. $x^2 + 4y^2 + z^2 + 4xy - 2xz - 4yz$
5. $6x^5 - 10x^4 - 8x^3 + 18x^2 - 8x + 4$
6. $x^5 - 81x$
7. $8x^4 - 8x^3y + 2x^2y^2 - 3xy^3 + y^4$
8. $81x^4 - 108x^3 + 54x^2 - 12x + 1$
9. $a^4 + a^2b^2 + b^4$
10. $3x^6 - 11x^4 + 19x^2 - 15$
11. $4x^4 + 12x^3 - 7x^2 - 24x + 16$
12. $x^2 + 2xz + z^2 - y^2$
13. $15x^4 - 16x^3 + 16x^2 - 25x + 14$
14. $5x^2 + 6x - 11$
15. $t = 6$ seconds (Factors: $-16(t - 6)(t + 1)$)
16. $V(x) = 4x^3 - 80x^2 + 384x$
17. Length = $3x - 2$
18. Original side length = 3 units
19. $P(x) = -2x^2 + 120x - 400$; Max profit = \$1400 at $x = 30$
20. Area = $4x^2 + 100x + 600$
21. $(2x - 1)(2x + 1)(2x + 3) - 15(6x + 3) = 80 \Rightarrow 8x^3 + 12x^2 - 92x - 128 = 0$
22. $2x^2 - x - 1$
23. $A(x) = 6\pi x^2 + 14\pi x + 4\pi$
24. Max height = 9 units, Base width = 6 units
25. $x^2 - x - 2$
26. $x^3 + 2x^2 + 4x + 8$
27. $x^2 + 3x - 2$
28. $3x^2 + 4x + 5$
29. $x^4 - 2x^3 + 4x^2 - 8x + 16$
30. $2x^2 - 2x - 2$
31. $3x^2 - 4x + 2$
32. $x^2 - 3x + 2$
33. $x^2 + 3x - 4$
34. $k = 1$
35. $3x^2 - 2x - 2$
36. Quotients: $x^2 - 2x$, $x^2 - 4$, $x^2 + 2x$ respectively
37. $x^2 - 3x + 2$
38. $4x^2 - 6x + 9$
39. $3x^2 - 3x + 4$
40. Quotient = $x^2 - x$; Remainder = 0